

# Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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## Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. That grain is the uniform bound's, not phase order's: three results replace it by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states

glue to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed  $2D/\kappa$ , for a  $\kappa$  derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

# 1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter

14 two relations as formal correlates of conscious unity and a minimal self; their  
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information  
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),  
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene  
19 et al., 2014), and predictive-processing accounts derive both from a hierar-  
20 chical generative model (Friston, 2010). Here the three requirements are  
21 kept apart rather than derived from one another, so that each carries its own  
22 assumptions and admits its own measurement; Section 10.3 returns to the  
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a  
25 description of the hand’s relation to it can agree about where the cup is while  
26 contributing different contents. Synchronizing the two processes would leave  
27 disagreement about that shared quantity unresolved. An internal representa-  
28 tion must additionally locate the perceiver within the represented situation.  
29 This example motivates the requirements; the mathematical constructions  
30 use spatial measures and fields in real normed vector spaces, whose identifi-  
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these  
33 requirements and the assumptions each connection needs. A declared ac-  
34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on  
35 overlaps convert phase order into a bound on content disagreement. Compat-  
36 ible spatial measures have a unique global extension; approximate agreement

controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold  $K_c = 2D$  for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition makes coupling a purchased quantity: a declared actuator's stored energy bounds the mean coupling it supplies, so coherence on the stationary branch carries a resource requirement in units a measurement can reach, and for cortex that requirement reduces to a figure the mode prices must deliver. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps it from

60 being vacuous in a large population. A stability criterion on the reweighted  
61 coupling kernel, together with an existence condition on an amplitude layer,  
62 settles when a low resultant reports weak coupling and when it reports a  
63 winding field. And the composition is machine-checked together with a wit-  
64 ness discharging all eight hypotheses at once, which is what makes their  
65 conjunction non-empty rather than a list no object satisfies.

66     Granting those premises, collected in Table 1, the account runs as follows.  
67 Extracellular contraction on awakening raises endogenous field amplitude  
68 and with it the effective coupling between cortical oscillators. Once that  
69 coupling passes  $2D$ , a coherent phase branch exists, and from an initial spread  
70 within a quarter turn the population settles onto it; other initial data leave  
71 windings stationary at the same coupling. Encoders shared where cortical  
72 territories overlap turn that phase order into agreement about the quantities  
73 the territories describe in common, so the local descriptions glue: there is  
74 one state of which each is a restriction. Part of that state encodes the state  
75 itself, and a readout of that part reconstructs it accurately enough, and  
76 sensitively enough across the states that matter, to make it the unique state  
77 the system models itself as being. On this account the glued state is what  
78 makes distributed activity one situation, and its accurate self-reconstruction  
79 is what gives that situation a perspective. Every step of it is conditional.

80     Section 2 collects the results that hold whatever physical field is proposed  
81 and whatever substrate carries it; a reader who rejects the cortical hypothesis  
82 of Section 9 still has to accommodate them. Sections 3–5 then develop the

three requirements together (Fig. 1), and Section 6 reads the unconditional results against a machine candidate. Section 7 composes them, stating the thermodynamic constraints and all eight connecting hypotheses per branch; its supporting models of feedback, learning and resource supply have distinct state spaces and assumptions. Section 8 then examines what the observables can establish, before Section 9 develops the cortical hypothesis and an awakening experiment. The hypothesis is that extracellular contraction increases field-mediated coupling; its direction, rate and relation to recovery require independent calibration. Supplemental Material organizes the constructions and controls behind the claims they support. Table 2, at the end, lists the symbols and the section that defines each.

## 2 What holds without the cortical hypothesis

Most of what follows is conditional. Table 1 names eight hypotheses that the composition consumes rather than derives, and the cortical identification of Section 9 is a further commitment on top of them. Five results are not conditional in that way. They constrain any account that reads unity off phase coherence, whatever physical field is proposed to carry the coupling and whatever substrate carries the field. Each is stated below where it is used; collected here, they are the article's fixed points.

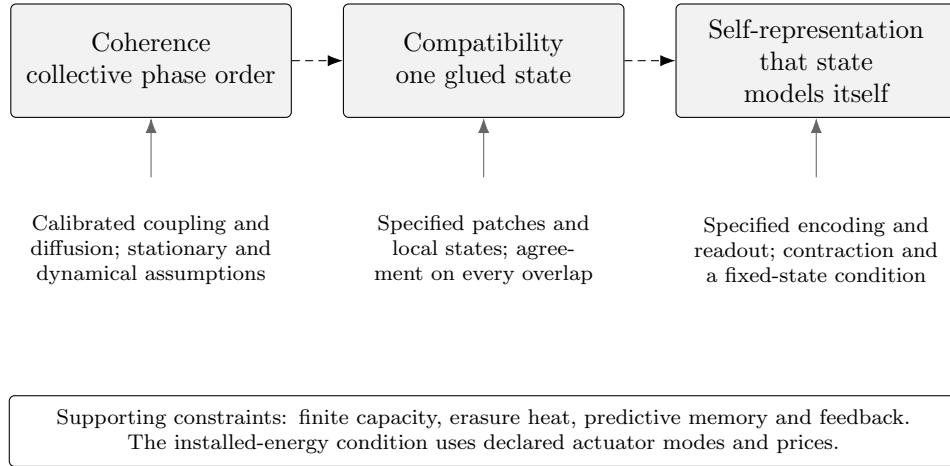


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

102 **Phase order constrains content only inside a patch too small to**  
 103 **carry unity.** Assign every patch the same phase and let two local mass  
 104 profiles disagree at a shared site. The profiles are chosen independently of  
 105 phase, so synchrony places no constraint on their disagreement, and they  
 106 have no common extension: perfect locking is consistent with local descrip-  
 107 tions that do not glue (Section 4). What the positive result of Section 4.2  
 108 costs is exactly this gap. A bound on content disagreement needs encoders  
 109 already related on overlaps, and that relation is a supplied fact about the de-  
 110 scriptions rather than a consequence of the dynamics. Supplying it relocates  
 111 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord

112 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that  
 113 uniform reading carries information only at 3 sites or fewer. The grain is the  
 114 uniform reading's and not phase order's, and Section 4.2 states three results  
 115 without it: a scale-free count of the pairs that can disagree, which names  
 116 none of them; a bound chained through the sites a cover's patches share,  
 117 which travels a distance linear in the hops taken; and, at a frequency-locked  
 118 configuration, a bound read off the coupling graph's algebraic connectivity,  
 119 which carries no population count at all. What none of the three changes is  
 120 the constraint stated here: a count locates no site, and the other two con-  
 121 sume a declared cover or declared hardware data, so phase order by itself  
 122 still says nothing about territories too far apart to share a patch, which is  
 123 where the compatibility claim is made.

124 **The order parameter does not determine the state.** A field wind-  
 125 ing once across a periodic sheet has resultant  $r = 0$  — the order parameter  
 126 of Eq. (1) below — while every patch of it stays locked, and this is not a  
 127 contrived initial condition. On a periodic array of any dimension, a winding  
 128 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-  
 129 cause the coupling term is odd in the separation against a kernel even in it  
 130 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-  
 131 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,  
 132 diffusion and kernel. A low resultant therefore reports either weak coupling  
 133 or a winding field, and no measurement in  $r$  alone separates them. The state

134 also carries an integer the resultant does not: read around a ring, the same  
135 winding has winding number  $q$ , and a nonzero  $q$  obstructs every global real-  
136 valued phase lift (`winding_degree_obstructs`). The resultant is an average  
137 over sites and moves when they are reweighted or the patches redrawn; the  
138 degree telescopes and does not.

139 **A unique fixed point is not a self-model.** A constant readout con-  
140 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness  
141 is consequently the wrong criterion, and Section 5.2 gives the right shape  
142 of one: reconstructing  $M$  states pairwise more than  $2\varepsilon$  apart within error  $\varepsilon$   
143 requires  $M$  distinguishable codes, so an alphabet of  $2^b$  codes caps the family  
144 at  $M \leq 2^b$ . This is a counting argument and holds for any encoder and  
145 readout whatever.

146 **Where the encoding is a restriction, the region must resolve the**  
147 **family.** Read the code as the state's own restriction to a region and  
148 the criterion moves off the readout entirely. A boundary whose region  
149 reads two relevant states identically cannot reconstruct them more than  $2\varepsilon$   
150 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant  
151 boundary that does reconstruct a separated family restricts injectively on  
152 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the  
153 region sits is fixed before an encoding is chosen, so no readout supplies  
154 resolution the region lacks.

155 **Agreement after a change takes communication.** In a synchronous  
156 network whose updates read only the site’s own state and its declared neigh-  
157 bours’ messages, two configurations agreeing on a site’s causal past through  
158  $T$  rounds leave that site identical at round  $T$ . Two interventions outside  
159 that past that require reports more than  $2\varepsilon$  apart therefore cannot both be  
160 reported from it within  $\varepsilon$  by the deadline (Section 5). Snapshot compatibil-  
161 ity and timely compatibility are therefore different requirements, and cutting  
162 the path removes the second while leaving the first intact.

163 None of these depends on the coupling being electromagnetic, on the sub-  
164 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6  
165 applies four of them — all but the one about the resultant, which is a state-  
166 ment about a spatial observable — to a candidate that is neither biological  
167 nor field-coupled.

## 168 **3 Coherence: physical coupling and its re-** 169 **source condition**

### 170 **3.1 Dynamics and the stationary threshold**

171 The finite stochastic model is

$$d\theta_i = \left[ \omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

172 Here  $\theta_i$  is the phase of oscillator  $i$ ,  $\omega_i$  its natural frequency,  $D > 0$  is phase dif-  
 173 fusion,  $W_i$  are independent Wiener processes, and  $K_{ij}$  has inverse-time units,  
 174 with angles dimensionless. Uniform mean-field coupling is  $K_{ij} = K/N$ ; fre-  
 175 quency heterogeneity and diffusion are distinct parameters. The stationary  
 176 calculation below uses identical frequencies and this scalar coupling. Ap-  
 177 plying it to a spatial kernel requires an aggregation rule and a check of the  
 178 resulting approximation.

179 For identical frequencies in their rotating frame, the positive normalized  
 180 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

181 where  $I_n$  are modified Bessel functions and  $\psi$  is the collective phase. The for-  
 182 mal derivation uses the absence of stationary current for a periodic gradient  
 183 drift to obtain this density, then verifies normalization and self-consistency.  
 184 At  $K \leq K_c = 2D$ , only  $r = 0$  is self-consistent (Sakaguchi, 1988). Above  
 185 threshold, exactly one positive solution coexists with zero and increases  
 186 strictly with  $K$ . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

187 Figure 2 shows this stationary branch.

188 Linearization about the uniform density gives rate  $K/2 - D$  for the first  
 189 harmonic and  $-Dn^2$  for harmonics  $n \geq 2$ . A growing mode therefore exists

190 exactly above threshold. About the coherent density, the quadratic form  
 191 is strictly dissipative transverse to collective rotation, whose rate is zero.  
 192 These linear estimates concern the limiting equation. Relating them to finite  
 193 stochastic trajectories requires a dynamical mean-field limit and branch selec-  
 194 tion; exchange symmetry alone does not establish the needed independence  
 195 (Sznitman, 1991).

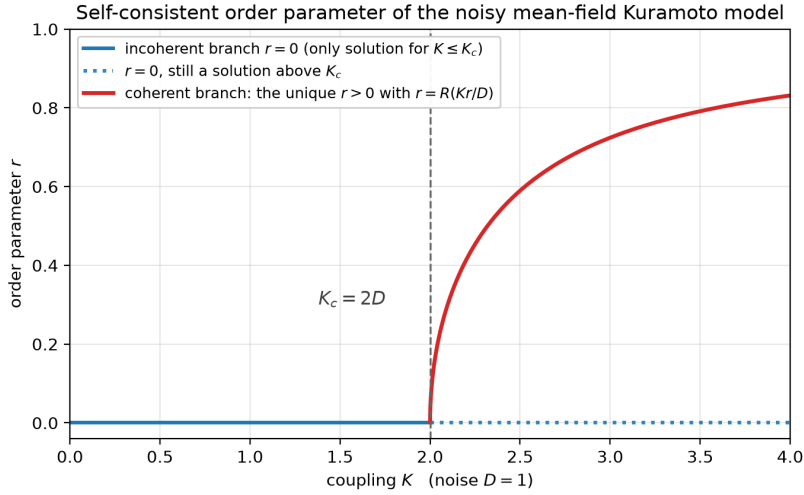


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion  $D = 1$ . The positive coherent branch appears continuously above  $K_c = 2D$  and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

### 196 **3.2 Installed energy bounds mean coupling**

197 A spatial model replaces the finite sum by integration against a kernel. One  
 198 declared implementation carries response modes with continuous profiles  $\phi_i$ ,

199 installation prices  $u_i$  and nonnegative occupancies  $c_i(\Xi)$  in configuration  $\Xi$ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

200 The same configuration sets the reciprocal kernel and stored energy. An  
 201 executed transition changes the kernel through its changed occupancies and  
 202 requires work  $U(\Xi') - U(\Xi) + Q$ , with  $Q$  positive into the reservoir. The  
 203 installed kernel is averaged over the resulting configuration law  $p(\Xi)$ . Cell-  
 204 pair reconstruction on refining spatial partitions converges to its product-  
 205 measure integral under the specified compactness and sampling assumptions;  
 206 the refinement index is independent of the number of control updates.

207 Normalize the substrate measure  $\mu$  to one. Write  $m_i = \int \phi_i d\mu$  and  
 208 declare  $\kappa > 0$  with  $m_i^2 \leq \kappa u_i$  for every mode. Factorizing each mode's  
 209 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

210 If this mean coupling is the  $K$  of the scalar model,  $\kappa U_{\text{inst}} \leq 2D$  excludes a  
 211 coherent stationary branch and any growing Fourier mode about the uniform  
 212 density. Thus coherence on that branch requires  $U_{\text{inst}} > 2D/\kappa$ . The condition  
 213 is necessary: energy above it need not produce sufficient coupling because  
 214 the bound can be loose.

215 The quantity constrained is installed energy rather than dissipated heat,  
 216 and the condition binds through  $\kappa$  rather than through the energy, since  
 217 any macroscopic substrate carries installed energy far above the thermal  
 218 floor. Read forwards it therefore excludes no macroscopic candidate. Read  
 219 backwards it is a number. Take the region the field arithmetic takes, a  
 220 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates  
 221 Supplemental Material collects and over one phase-diffusion time  $1/D = 2$  s  
 222 at  $D = 0.5$  rad<sup>2</sup>/s: the installed energy is  $1.0 \times 10^{-6}$  J, and the condition is  
 223 met at every  $\kappa$  above  $9.6 \times 10^5$  J<sup>-1</sup>s<sup>-1</sup>. That figure is what an identification  
 224 of cortical mode prices has to deliver, and the first identification delivering  
 225 less refutes the coupling-gated account for cortex at this region and interval.  
 226 Another interval rescales the figure by its ratio to this one.

227 What the condition discriminates, it discriminates through that factor.  
 228 The required  $\kappa$  scales as  $D^2$  and inversely with the region's cell count, so at  
 229 fixed mode prices the condition ranks candidate substrates by their installed  
 230 energy. In a declared implementation it decides outright, the learning witness  
 231 of Section 7 clearing the line after one update while its matched control stays  
 232 on the exclusion side at the same conversion factor and diffusion. What is  
 233 missing for cortex is a price list rather than a mechanism. A mode is a priced  
 234 rank-one contribution to the kernel, cortical field generation has no canonical  
 235 decomposition into such modes, and choosing the basis would choose the  
 236 answer. Applying the scalar threshold to heterogeneous or signed spatial  
 237 dynamics requires a justified reduction, and identifying the installed kernel

238 with a cortical field is a separate physical commitment.

## 239 4 Compatibility: from coordinated activity 240 to a global description

### 241 4.1 Why common timing is insufficient

242 For overlapping patches  $U$  and  $V$ , compatibility means that restricting either  
243 local description to  $U \cap V$  gives the same object. Descriptions can agree on  
244 shared quantities while contributing different information elsewhere. Com-  
245 patible sections of the specified sheaf have a unique global extension. On  
246 a finite cover, local finite spatial measures agreeing as measures on every  
247 overlap also have a unique finite-measure extension whose sheaf image is the  
248 glued section. No mass-one normalization is imposed.

249 In the cup example, visual and reaching descriptions must express a  
250 shared quantity, such as cup location, in common coordinates. Restriction  
251 retains that quantity while dropping details particular to each description.  
252 Agreement permits surface appearance and hand configuration to contribute  
253 to one situation. The mathematical counterexample assigns the same phase  
254 to every patch while two local mass profiles disagree on a shared site. Those  
255 profiles have no common extension. Because they are chosen independently  
256 of phase, synchrony places no constraint on their disagreement.

257 This distinction limits the biological interpretation. Receptive territories

258 and content domains must be related to spatial patches, and the descriptions  
 259 must support the required extension. Matching overlap marginals of de-  
 260 coded probability laws need not determine a unique global joint distribution  
 261 (Abramsky & Brandenburger, 2011).

262 The cover is itself a choice, and the construction constrains it in one way  
 263 only: synchronized patches agree where they overlap. Where they do not  
 264 overlap there is nothing to compare, so a cover by pairwise disjoint patches  
 265 satisfies that obligation for *every* assignment of local descriptions and phases  
 266 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-  
 267 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-  
 268 ship of the class then reports nothing about the system beyond the patches  
 269 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy  
 270 the theorem does not assume: that distinct patches meet (`HasNonempty-`  
 271 `Overlaps`), and that what they share is a quantity fixed independently of  
 272 the agreement it is being used to establish.

## 273 4.2 When coherence does constrain content

274 A declared relation between phase and content gives a positive result. Let  
 275 contents take values in a real normed vector space  $\mathcal{C}$ . Patch  $i$  reads an  
 276 assigned oscillator through  $e_{i,x}$  at site  $x$ , evaluated on the circle point  
 277  $(\cos \theta, \sin \theta)$ . Assume a uniform Lipschitz constant  $L$  in chord distance and  
 278 require  $e_{i,x} = e_{j,x}$  wherever patches  $i$  and  $j$  overlap. Encoders may vary across  
 279 sites; their equality on overlaps is a supplied relation between descriptions of

280 the shared quantity. The order-parameter identity  $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$   
 281 implies  $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$  for each pair. Consequently every  
 282 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

283 Here discrepancy is the norm of the difference, and  $r$  and  $N$  are read on  
 284 whichever population the bound is applied to. That choice is what decides  
 285 whether the estimate says anything. A patch carries an order parameter of  
 286 its own, and reading the bound there carries that patch's resultant and that  
 287 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`  
 288 `coherence`), so a large population observed through small patches is con-  
 289 strained by the small count. Read instead on the whole array, the factor  $N$   
 290 makes the estimate weak, because it draws a pointwise guarantee from an  
 291 average over  $N^2$  pairs. At  $r = 1$  the observations agree exactly at either  
 292 scale. A shared scalar encoder is a special case. Dropping the overlap rela-  
 293 tion permits disagreement even at perfect locking. A common nonexpansive  
 294 linear projection preserves agreement, but agreement of a projection does  
 295 not establish agreement of the full vector.

296 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate  
 297 constrains anything only where  $\sqrt{2} N\sqrt{1 - r^2}$  falls below that. The winding  
 298 sweep of Section 8.3 fixes both ends of that range on one sheet. Read on the  
 299 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is  $382.5 L$

300 and constrains nothing; holding that order fixed, the count alone brings it  
 301 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair  
 302 of the same winding, the smallest patch there is, gives  $0.1388 L$ . Shrinking  
 303 patches therefore thins the overlaps at the same rate as it tightens the bound,  
 304 and the two ends are not reached by the same cover. At a columnar grain  
 305 the estimate constrains agreement inside one locally synchronous population  
 306 and is vacuous across the distant territories whose agreement unity is about.

307 That vacuity is the uniform bound's, and three results decline the step  
 308 that puts  $N$  there: a sum of  $N^2$  nonnegative terms bounded by its total,  
 309 which permits all the disorder in a population to sit in the pair the conclu-  
 310 sion is about. Asking instead how *many* pairs can be far apart is Markov's  
 311 inequality on the same sum, and its answer carries no population count: at  
 312 order  $r$  the fraction of pairs separated by a chord of  $c$  or more is at most  
 313  $2(1 - r^2)/c^2$  at any size of population (`chord_fraction_le`). The same sum  
 314 read by rows bounds the fraction of *sites* that are far from a fixed fraction  
 315 of the population (`site_fraction_le`). Read on the patch above, where the  
 316 uniform bound gives  $382.5 L$  and constrains nothing, this gives a fraction  
 317 at most 0.37 of pairs disagreeing by more than  $L$ , and it improves like the  
 318 square of what the uniform bound improves like. It is a different quantifier  
 319 and not a stronger statement: a count names no pair, the exceptional set  
 320 it permits is unlocated and may be exactly the sites a cover's overlaps sit  
 321 on, and a guarantee about a named overlap therefore still needs the uniform  
 322 bound at its own grain.

323 Distance is reached by chaining rather than by counting. Index the cover  
 324 by its nerve, in which two patches are adjacent when they share a site.  
 325 The chord is a metric, so a bound  $\beta$  on the separation inside every patch  
 326 composes along a walk: a walk of  $k$  hops bounds its endpoints by  $(k + 1)\beta$   
 327 (`chord_le_of_patch_walk_coherence`). The count is  $k + 1$  and not  $k$  because  
 328 both endpoint sites pay a step inside their own patch while the shared sites  
 329 are interior to the chain, and the content statement follows for any cover  
 330 whose nerve has diameter  $k$  (`compatible_of_patch_nerve`). No patch need  
 331 contain both sites, which is the situation of every pair of territories far enough  
 332 apart, so shrinking patches and spanning territories are two ends of one chain  
 333 rather than two ends of one bound. What it costs is that the accumulation is  
 334 linear and nothing makes it sublinear: at the sheet's nearest-neighbour value  
 335  $\beta = 0.1388 L$  the chain stays inside the maximum chord for 13 hops, which  
 336 is 14 patch diameters and 0.219 mm of that sheet. The territories whose  
 337 agreement the compatibility claim is about are separated by centimetres,  
 338 which is two orders of magnitude beyond where this chain saturates; the route  
 339 reaches further only on a cover whose patches are larger or more coherent,  
 340 and both are measurements. Connectivity of the nerve is a property of the  
 341 cover, supplied rather than derived, and a cover whose nerve is disconnected  
 342 supports no conclusion between its components.

343 The third result reads no resultant at all. Where a configuration is  
 344 frequency-locked to a common rotation  $\Omega$ , its phase differences are not free:  
 345 each site's detuning is balanced by its coupling term,  $\omega_i - \Omega = \sum_j K_{ij} \sin(\theta_i -$

$\theta_j$ ), which is the Kuramoto equations at a rigidly rotating solution rather  
 than a further postulate. Pairing that balance with the configuration itself,  
 symmetrizing, and applying Jordan's inequality termwise turns it into the  
 coupling graph's Dirichlet form, whose algebraic connectivity  $\lambda_2$  then bounds  
 the spread about the mean phase  $\bar{\theta}$  by  $2\lambda_2\|\theta - \bar{\theta}\| \leq \pi\|\omega - \Omega\|$  (`spread_le_`  
`of_frequency_locked`). Read at a named pair it gives a content residual  
 of  $L\pi\|\omega - \Omega\|/(\lambda_2\sqrt{2})$  with no population count, no patch and no cover ge-  
 ometry in it (`chord_le_of_frequency_locked`, `compatible_of_frequency_`  
`locked`), and it improves as the coupling graph becomes better connected,  
 which is the dependence a resultant cannot express. Where it is informative  
 is fixed by the same hypothesis that licenses it: confinement to a quarter  
 turn already bounds every chord by  $\sqrt{2}$  with no coupling data at all, so the  
 spectral reading adds something exactly where  $\lambda_2 > \pi\|\omega - \Omega\|/2$ . Four things  
 are declared and none is discharged:  $\lambda_2$  is hardware data in the standing of  $\kappa$ ;  
 the locked configuration is assumed, and nothing here proves one exists; the  
 quarter-turn confinement is assumed and is not implied by locking; and the  
 detuning enters in the population's  $\ell^2$  norm, so a population whose frequency  
 spread grows with its size pays for that growth through the data. What is  
 removed is the unconditional factor, not the physics of heterogeneity.

Structural connectivity is a declared graph and is what supplies  $\lambda_2$ . Mea-  
 sured functional connectivity supplies it no better: it is an estimate, so it  
 inherits the decoder failure modes of Section 8, a shared prior shrinking esti-  
 mates toward agreement and regional bias manufacturing disagreement, and

369 feeding an estimate to a hypothesis strengthens no conclusion drawn from  
 370 it. Using it to choose the cover is worse than neutral, being the move of  
 371 Section 4 that empties a gluing claim, in different clothes.

372 Mix retained fields with observations at gain  $\eta \in [0, 1]$ :  $s_i \mapsto (1 - \eta)s_i +$   
 373  $\eta o_i$ . Content and observation discrepancies bounded by  $\varepsilon$  and  $\delta$  give a next  
 374 discrepancy at most  $(1 - \eta)\varepsilon + \eta\delta$ . Repeated observations bounded by  $\delta$   
 375 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

376 Thus agreeing observations preserve exact compatibility and, for  $\eta > 0$ , re-  
 377 duce an initial mismatch. Approximate observations leave a nondecaying  
 378 term in the error bound. Taking a uniform bound on Eq. (6) connects this  
 379 update to the phase process.

### 380 4.3 What approximate agreement determines

381 Exact gluing requires exact restrictions. For nonnegative profiles on a finite  
 382 substrate, suppose every overlap discrepancy is at most  $\varepsilon$  in uniform distance.  
 383 Weights that vanish outside their patches and sum to one at each site select  
 384 a global profile by weighted averaging. The selection lies within  $\varepsilon$  of each  
 385 patch there, and any two such selections differ by at most  $\varepsilon$ , independently  
 386 of cover multiplicity. An  $L$ -Lipschitz readout consequently varies by at most  
 387  $L\varepsilon$  across selections.

388 At zero tolerance, the selection is the unique exact extension. At every

389 positive tolerance there are families with no exact extension; selection then  
390 gives an approximation whose dependence on weights is bounded. Supple-  
391 mental Material gives a three-site witness attaining the bound. The mass-  
392 profile construction uses the scalar case of the content dynamics and addi-  
393 tionally requires nonnegativity of the initial profiles and observations. The  
394 general vector result supplies no identification with the spatial-measure sheaf.

395 These constructions concern declared objects: a specified cover, contents  
396 valued in a normed space, and encoders related on overlaps. They select  
397 neither the neural cover nor the decoder, and identifying their contents with  
398 perceptual content, or establishing that the identification bears on experi-  
399 ence, is a further modelling task; Section 8 sets out what a measurement of  
400 it would have to establish.

## 401 **5 Self-representation: a state that models it-** 402 **self**

### 403 **5.1 The reflexive construction**

404 Compatibility gives a global state. Reflexivity relates it to an encoding within  
405 that state. Equip global sections with a complete metric. If the additional  
406 restriction-resonance condition identifies encoding with restriction to a region

407  $\mathcal{A}$ , the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

408 The region and readout are fixed across the state space. If  $F$  contracts and  
 409 fixes the section  $s_*$  formed from the local descriptions, Banach’s theorem  
 410 makes  $s_*$  its unique fixed point (Banach, 1922). The formal bridge E89 sup-  
 411 plies the contraction and  $F(s_*) = s_*$ ; restriction resonance is separate. Su-  
 412 percritical coupling makes the stipulated contraction factor  $\Lambda = e^{-(K-2D)\tau/2}$   
 413 less than one for a relaxation time  $\tau > 0$ . Constructing a map with that  
 414 factor remains a modelling obligation.

415 In the cup example, the global state includes the represented scene and the  
 416 perceiver’s relation to it. A local encoding must retain enough information  
 417 for the readout to recover that modelled state; cup position alone would be  
 418 insufficient if hand configuration and viewpoint could not be reconstructed.

419 For the same map with Lipschitz factor  $\Lambda < 1$ , approximate reconstruc-  
 420 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

421 At the stipulated rate this radius grows as  $2\varepsilon/((K-2D)\tau)$  near threshold, for  
 422 fixed positive  $\varepsilon$  and  $\tau$ . Exact uniqueness survives while the error guarantee  
 423 weakens.

## 424 5.2 Reconstruction across a family of states

425 Sensitivity can be required quantitatively. Declare relevant macrostates, an  
 426 encoder  $E$  and readout  $R$ , and require  $d(s, R(E(s))) \leq \varepsilon$  throughout that  
 427 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

428 Thus  $M$  relevant states pairwise more than  $2\varepsilon$  apart require at least  $M$   
 429 distinguishable codes; an alphabet of at most  $2^b$  codes requires  $M \leq 2^b$ .  
 430 Applied to the reflexive boundary,  $E$  is its local encoding and  $R \circ E$  is the  
 431 existing self-prediction map.

432 This is an additional accuracy criterion on a family, with no requirement  
 433 for multiple exact fixed points. A blind encoder or constant readout fails it  
 434 for separated states despite having a unique fixed point.

435 Where the encoding is read as the state's own restriction to a region, the  
 436 criterion becomes a condition on the region. Restriction resonance requires  
 437 the encoder and that restriction to identify the same pairs of states, so a  
 438 boundary reconstructing two relevant states more than  $2\varepsilon$  apart while its re-  
 439 gion reads them identically is not resonant (`not_isRestrictionResonance_`  
 440 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs  
 441 a pairwise separated family restricts injectively on it (`restrictToAvatar_`  
 442 `injOn_of_isRestrictionResonance`). Which states a region reads identi-  
 443 cally is fixed by where the region sits, before an encoding is chosen, so the

444 resolution the family demands is the region's and no readout supplies it. The  
 445 implication runs one way: a region that does resolve the family is not thereby  
 446 shown to carry a resonant encoding, and E89 supplies no such resolution.

447 Those two demands are jointly satisfiable with nothing folded anywhere.  
 448 Take the region to be the whole substrate: restriction is then the identity, so  
 449 the boundary is resonant, reconstructs every declarable family exactly, and  
 450 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A  
 451 properness condition on the region (`IsProperAvatar`) is therefore a separate  
 452 requirement rather than a consequence of resonance and resolution, and what  
 453 a fold into a local region is asked for. Its quantitative form replaces the  
 454 alphabet by a metric. With an  $L$ -Lipschitz readout, a family reconstructed  
 455 within  $\varepsilon$  whose members lie pairwise more than  $d_{\min}$  apart has  $\delta$ -separated  
 456 codes whenever  $\delta L \leq d_{\min} - 2\varepsilon$ , so the family is no larger than the  $\delta$ -packing  
 457 number of what the region carries (`encard_le_packingNumber_restrict-`  
 458 `ToAvatar`). That is the shape a continuous family of states needs, and at  
 459 the whole substrate it says nothing, because the region's range is everything.  
 460 The gain  $L$  is declared and derived here for no physical readout, and at  
 461 unbounded gain the bound says nothing.

### 462 **5.3 Agreement after a change takes communication**

463 Snapshot compatibility does not specify how quickly descriptions respond  
 464 when a shared quantity changes. In a synchronous network whose updates  
 465 read only the site's own state and declared neighbours' messages, two initial

466 configurations agreeing on a site’s causal past through  $T$  rounds produce the  
467 same state there at round  $T$ . The deadline consequence is the one Section 2  
468 states, and it follows from the reconstruction bound in Section 5.2: a report  
469 from that site is a readout of a code that the two interventions leave identical.

470 A three-site delay line witnesses failure after one round and exact report-  
471 ing after two; cutting the path prevents delivery at every round. A report  
472 may still happen to be correct about one fixed world. The bound concerns  
473 responsiveness across interventions.

474 Throughout this section *prediction* names a reconstruction map, and the  
475 relevant family of states, the metric, the tolerance, the communication graph  
476 and the deadline are all declared inputs, as are the represented variables and  
477 the fidelity counted as accurate. Reading the map as temporal neural pre-  
478 diction would require dynamics; reading the resulting code count as physical  
479 memory capacity would require a hardware model, which a state-dependent  
480 readout would have to extend.

## 481 **6 What the conditions constrain in a digital** 482 **candidate**

483 The results of Section 2 hold for any candidate, and for a large language  
484 model (LLM) running on graphics processing units (GPUs) four of them are  
485 checkable rather than merely stateable. Those four are where this framework  
486 bears on claims made for such systems. It does not exclude them, and no re-

487 sult here could; what it does is fix what a claim of unity or self-representation  
488 commits its maker to measuring.

489 **Codes, not reports.** Reconstruction across a family is a counting con-  
490 straint before it is anything else. If two macrostates the candidate is said  
491 to represent satisfy  $E(s) = E(t)$  while  $d(s, t) > 2\varepsilon$ , Eq. (10) rules out re-  
492 constructing both within  $\varepsilon$ , whatever the readout. A system’s reports are  
493 readouts. Identical verbal reports across two conditions are therefore consis-  
494 tent with identical internal codes and settle nothing about the states. The  
495 converse claim—that the system represents its own condition across a de-  
496 clared family—asserts that its internal code separates that family, which is a  
497 statement about activations under intervention and not about text. Naming  
498 the channel makes the count concrete. Where the code is the token sequence  
499 a model emits—which is what its next step reads—the alphabet is the de-  
500 ployed vocabulary (`tokenChannel`), and a family reconstructed through it  
501 is no larger than the vocabulary size raised to the number of steps (`card_`  
502 `le_card_tokens`). Over an extended exchange that number is large and the  
503 bound slack. At a single step it is one vocabulary’s worth, so claiming that a  
504 single emission reports the model’s own condition across a declared family as-  
505 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim  
506 resting on activations rather than on text has declared a different channel,  
507 to which the same count applies with that channel’s alphabet. The measure-  
508 ment follows from the statement: declare the family and the fidelity first,

509 then test whether the encoding region distinguishes its members.

510 **The region, not the readout.** Where the encoding is read as the state's  
511 restriction to a region of the machine, the demand falls on the region before  
512 any readout exists. For a decoder-only transformer the causal mask makes  
513 the network an instance of the update model of Section 5: the activation  
514 at position  $t$  and layer  $\ell$  reads positions at most  $t$  at layer  $\ell - 1$  and nothing else. A region's causal past is therefore fixed by the architecture rather  
515 than chosen by interpretation. That instance is formal, and sharper than  
516 slow growth: the causal past of position  $t$  stays inside the positions at most  
517  $t$  at every depth (`mask_ball_subset_1e`). The deadline bound of Section 5  
518 therefore fires at every deadline at once rather than at a chosen one, its communication graph given rather than assumed (`mask_no_guarantee`). That is  
519 a statement about one forward pass; across steps a model does read its own  
520 prior output, which is the channel the preceding paragraph counts. Read  
521 against the coupling proposed here the asymmetry runs both ways, and neither way is an advantage. A kernel of full support has no neighbourhood  
522 structure for the bound to bite on: after one round the causal past is the  
523 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable  
524 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physical  
525 field propagates at a finite speed, so treating one that way is a calibration  
526 against the phase time scale and not a theorem. What the comparison does  
527 fix is where the bound has force. Not on the attention graph, where a posi-

531 tion reads every position at or before it and the fan-in is the context length.  
 532 On the physical interconnect, fan-in is genuinely bounded: a graph whose  
 533 sites each hear from at most  $\nu$  others reaches at most  $(T + 1)\nu^T$  sites in  
 534  $T$  rounds (`card_ball_le_mul_pow`), so a model sharded across many devices  
 535 pays a number of rounds growing like  $\log_\nu$  of the device count. Buying that  
 536 deadline back costs wiring: reaching every site in one round requires a device  
 537 hearing from all but one of the others (`le_degree_of_reaches_one`). Both  
 538  $\nu$  and the device count are declared, as the communication graph and the  
 539 deadline already are, and what the count bounds is the graph a guarantee  
 540 requires rather than what hardware is buildable. A claim that a model re-  
 541 ports on its own processing consequently carries an obligation to say which  
 542 positions and layers the report’s causal past contains. A report that appears  
 543 to track an intervention outside it indicates a confound or a different region,  
 544 not a faster channel.

545 **Agreement about what.** Unity requires the candidate’s local descriptions  
 546 to agree on independently identified shared quantities (Section 4). Common  
 547 timing, a shared residual stream and fluent output supply none of that, for  
 548 the reason Section 2 gives: coordination bounds content disagreement only  
 549 through encoders already related on overlaps. Which quantities two parts  
 550 of a model describe in common is a decoding question, carrying the failure  
 551 modes of Section 8. The cover is a choice, and choosing it to secure agreement  
 552 empties the claim.

553 **What is not excluded.** Digital representations can satisfy compatibility,  
 554 so finite hardware is compatible with the gluing construction and none of  
 555 the above is a no-go for machines. The coupling-gated route E56–E78 is a  
 556 different matter: it requires a calibrated physical coupling, its noise regime  
 557 and a connection from coordination to content agreement, and an attention  
 558 matrix or a shared clock supplies none of these identifications. Computing a  
 559 field model likewise leaves its realization in the computing substrate to be es-  
 560 tablished. Those physical obligations fall on the cortical proposal identically.  
 561 The capacity comparison those support comparisons lack is available, and it  
 562 favours neither side by construction. A continuum code space read at reso-  
 563 lution  $\delta$  holds at least its volume divided by the volume of a resolution cell,  
 564 so it out-resolves an alphabet of  $2^b$  codes exactly when that volume exceeds  
 565  $2^b$  cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold  
 566 the inequality runs the other way, and at a coarse enough resolution the re-  
 567 gion holds a single code. What decides the comparison is the volume, the  
 568 resolution and the dimension, all measured quantities, and continuity decides  
 569 nothing on its own: a continuum code space read at a finite resolution is a fi-  
 570 nite alphabet whose capacity is a bit count. For the same reason this account  
 571 credits an analog medium with no computation beyond a Turing machine’s.  
 572 Real-valued states carry unbounded capacity only at unbounded precision,  
 573 and the noise floor  $D > 0$  that Eq. (1) is stated against removes it.

574 The installed-energy condition of Section 3.2 is restricted in the same  
 575 way. It constrains a macroscopic candidate only through  $\kappa$ , which is hard-

ware data supplied by mode profiles and prices, and Supplemental Material reads the condition against a cortical metabolic budget to return the conversion factor it would require rather than a verdict. The hardware comparisons there have similarly restricted scope: fixed wiring can impose a correlation disadvantage, and point-supported kernels vanish under an atomless continuum measure, but neither comparison makes the representation conditions unattainable on GPUs. A universal exclusion would need a necessary condition for consciousness and a proof that every relevant implementation violates it, and the present composition supplies neither.

Whether experience requires a living substrate is the explicit subject of biological naturalism (Seth, 2026a,b). On that view the properties grounding experience are those of self-sustaining organisms, and simulating them does not secure them. The conditions stated here name no biological mechanism, but computation described as such does not satisfy them either: a candidate must exhibit a physically realized coupling in a measured noise regime, agreement about independently identified shared quantities, and an internal reconstruction meeting a declared fidelity. Whether a realization of those conditions that is not alive would be accompanied by experience is the point at issue, and these results do not settle it. A related position locates the difference in the mode of computation rather than in life as such: biological systems compute across scales that do not separate, and alongside discrete operations carry continuous-valued dynamics belonging to their physical medium (Milinkovic & Aru, 2026). The coupling condition stated

599 here is one measurable instance of that requirement, fixed to a physical field  
600 in a declared noise regime, and a description of such a field does not satisfy  
601 it.

602 The neuroscientific case against *human-like* consciousness in text-only  
603 LLMs concerns embodied sensory interaction, thalamocortical organization  
604 and biological regulation linking perception and action to survival (Aru et al.,  
605 2023). Their absence supports skepticism, conditional on their causal rele-  
606 vance to experience. Those mechanisms are not requirements of the abstract  
607 gluing and reconstruction definitions, and workspace-like representations re-  
608 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to  
609 investigate. What this framework adds to that investigation is an order for  
610 the questions: declare the family of states and the fidelity, fix the encoding  
611 region and read its causal past off the architecture, and only then ask what  
612 the system reconstructs. A claim that survives those three steps is a claim  
613 about the machine; one that does not survive them was a claim about its  
614 output.

## 615 **7 Thermodynamics and composition**

616 The phase and representation results concern coupling, content disagreement  
617 and reconstruction error. Thermodynamics constrains the physical opera-  
618 tions that could implement them, and it does so from the side rather than  
619 along the spine: no edge below derives coherence, compatibility or reconstruc-

tion from a heat bound, and the resource condition these edges terminate in  
is necessary without being restrictive for any macroscopic candidate (Sec-  
tion 3.2). What this half contributes is a ledger in which a resource claim  
is stated in checkable units. Finite-register capacity obeys  $H(p) \leq \log |X|$ ;  
erasure heat depends on specified system–bath assumptions (Landauer, 1961;  
Norton, 2005; Shenker, 2000). A separate boundary result concerns a field  
reaching disconnected vacuum components (Kibble, 1976). The composition  
connects these objects through the explicit hypotheses in Table 1.

## 7.1 Feedback changes the information constraint

For passive memory  $X_t$  driven by an environmental signal  $S_t$ , the predictive  
bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

The Markov structure  $X_t \rightarrow S_t \rightarrow S_{t+1}$  makes the bracket nonnegative (Still  
et al., 2012). Zero retained information also satisfies it, so a task and a  
learning rule require additional specification. Relating this bound to Lan-  
dauer heat requires a common thermal scale and budget, and memory waste  
bounded by erased entropy.

An acting agent changes the signal it will receive. Specify policy, world,

637 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

638 The second conditional term vanishes in the passive model. With feedback,  
639 the signed difference can be negative as action creates correlations with the  
640 future. A causal reading requires an environmental state sufficient to describe  
641 the relevant dynamics.

642 For a controlled substep, the active thermodynamic bound instead uses  
643 the joint entropy. Forward and reverse path laws give  $\Sigma = D_{KL}(P_F \| P_R) =$   
644  $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$  under local detailed balance, with thermal scale  
645  $k_B T > 0$  and heat positive into the reservoir (Horowitz and Esposito, 2014;  
646 Ito and Sagawa, 2013). Every forward path of positive probability must have  
647 a positive-probability reverse for this finite identity. A missing reverse gives  
648 infinite divergence. The supplement develops the support conditions and  
649 composes actuation with memory updates on their shared intermediate law.

## 650 **7.2 Learning and repeated operation need explicit re-** 651 **sources**

652 The finite examples distinguish selecting a feasible policy, relaxing a register  
653 toward an encoded reward, and learning an unknown task from observa-  
654 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 3.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy observable  $\mathcal{E}$ , thermal scale  $k_B T$ , local detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance  $B$  at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

### 7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with  $B$  the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one: strictly positive forward and reverse elementary paths, local detailed balance at one thermal scale, and entropy differences and heats of order  $k_B T$ , the witnesses being two-bit systems with interaction energies of that size. Cortex is not in it. The installed energy of Section 3.2 is  $2.4 \times 10^{14}$  times  $kT$  at body temperature over one region and one phase-diffusion time, and a budget  $B$

694 of that size stands against an entropy term of order  $k_B T$ , so Eq. (14) is  
 695 satisfied by any macroscopic candidate and excludes none of them. Reading  
 696 the register instead at the scale where its premises are physical, such as a  
 697 single channel, relocates the difficulty to E45Active, which would then have to  
 698 carry one molecular step’s budget to the convergence of a continuum kernel.  
 699 The active branch is an alternative route to the same node rather than a  
 700 stronger one, and no result of Section 3.2 passes through it: the installed-  
 701 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

| Connection                                           | Status               | Required assumption                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E12<br><i>capacity and boundary</i>                  | Physical<br>premise  | A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.                                                                                                                                                                                                                                                                                                                     |
| E23<br><i>boundary to register</i>                   | Bridge<br>assumption | The boundary is associated with a finite register whose update leaves states unreachable.                                                                                                                                                                                                                                                                                                                                               |
| E34 /<br>E34Active<br><i>register to dissipation</i> | Bridge<br>assumption | Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies. |

Table 1 (continued)

| Connection                                              | Status                  | Required assumption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E45 /<br>E45Active<br><i>couplings to<br/>continuum</i> | Modelling<br>assumption | Convergent spatial coupling energies.<br>Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| E56<br><i>field<br/>identification</i>                  | Physical<br>commitment  | A specified physical field realizes the limiting coupling and satisfies the kernel conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| E67<br><i>above<br/>threshold</i>                       | Physical<br>commitment  | That field operates above the synchronization threshold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| E78<br><i>coherence to<br/>cover</i>                    | Bridge<br>assumption    | Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity $\lambda_2$ instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of $\kappa$ . |

Table 1 (continued)

| Connection                                   | Status                  | Required assumption                                                                                                                                                                                                            |
|----------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E89<br><i>contraction to<br/>fixed point</i> | Modelling<br>assumption | A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$ . Reconstruction over a declared family imposes an additional code-separation criterion.                   |
| Installed<br>energy                          | Conditional<br>theorem  | Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$ . The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows. |

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

717 in one place and machine-checked together, and not that coherence follows  
718 from installed energy or self-representation from coherence.

719 Joint witnesses combine a register, boundary, refining mesh and small  
720 spatial cover; active witnesses add controlled thermal updates and a priced  
721 installation. They establish satisfiability under their stated identifications.  
722 Mesh convergence follows from refinement data, and the witnesses establish  
723 no common cortical mechanism. Supplemental Material maps the claims to  
724 their Lean declarations and records the counterexamples. The development  
725 declares no additional axioms beyond its libraries' foundational axioms.

## 726 8 What the observations establish

727 The composition holds under its premises. Whether observation can estab-  
728 lish those premises is a separate question, and the three requirements it  
729 connects concern different objects: phase dynamics, local descriptions and  
730 internal encodings. Their observables must retain these distinctions. The  
731 following controls show which mechanisms remain indistinguishable, which  
732 temporal approximations fail, which states one observable cannot separate,  
733 and what independent content measurements must establish. Each is a finite  
734 numerical regime on a declared model, with full parameters and controls in  
735 Supplemental Material, and none of them calibrates a cortical quantity.

## 736 8.1 Stationary distributions leave the mechanism un- 737 determined

738 Equation (2) gives a concentration–coherence relation with no fitted shape  
739 parameter. Estimate  $r$  from circular resultant length and  $a$  from density  
740 shape: inverting  $R$  from the measured  $r$  would make agreement automatic.  
741 Even independently estimated agreement identifies a statistical family shared  
742 by different mechanisms.

743 Let  $q(t)$  cross 2, let  $r(q)$  be its stationary branch, and fix  $D_0 > 0$ . Increas-  
744 ing coupling  $K(t) = D_0 q(t)$  at diffusion  $D_0$  and decreasing diffusion  $D(t) =$   
745  $K_0/q(t)$  at coupling  $K_0$  both give concentration  $a = q(t)r(q(t))$ . Uncoupled  
746 oscillators with prescribed drift  $-h(t) \sin(\theta - \psi)$ , where  $h(t) = D_0 q(t)r(q(t))$ ,  
747 and diffusion  $D_0$  have that same density. Their stationary distributions co-  
748 incide window by window despite different causes.

749 Resolved dynamics supply additional information. In the ideal model with  
750 continuous paths, quadratic variation accumulates at rate  $2D$  and conditional  
751 increment means constrain drift. Separating endogenous coupling from a  
752 matched common drive additionally requires a measured drive or controlled  
753 perturbation; identifying electromagnetic coupling requires evidence specific  
754 to that interaction. Supplemental Material gives the equivalence construction  
755 and its numerical check.

## 756 8.2 Finite ramps change the observable onset

757 The square-root law in Eq. (3) describes stationary solutions. A changing  
758 coupling requires time for the phase distribution to approach those solutions.  
759 In a deterministic mean-field control that crosses the threshold symmetrically,  
760 none of the scanned rates meets the declared tracking tolerance: slowing  
761 the crossing does not remove the terminal residual under that initialization.  
762 A slow control variable therefore supplies no general guarantee of branch  
763 tracking at onset. Supplemental Material gives the residual, rate range and  
764 admissible rates away from threshold.

765 The finite-population ramp study uses  $N = 2000$ ,  $D = 1$  and 32 replicas  
766 at 8 coupling speeds from  $v = 0.1$  to  $v = 0.0001$ . Escape is the first of three  
767 consecutive saved samples at  $r \geq 0.2$  after threshold. Faster ramps delay  
768 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The  
769 6 uncensored legs supply the delay scaling below; the decade-spaced ones  
770 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted  
771 onset exponents differ from those of the exact stationary branch evaluated  
772 on the same grids. At the fastest speed the ensemble mean never reaches the  
773 fit window, so its fitted slope does not estimate stationary onset.

774 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)  
775 makes the apparent exponent depend on ramp speed and fitting window.  
776 The specified deviation-based critical ramp speed is not bracketed by this  
777 sweep. With fluctuating coupling, the relevant stationary comparison also  
778 depends on rate: the mean coupling approximates order in the tested fast

779 regime, while averaging stationary order over coupling works for the tested  
780 slow drives confined to one side of threshold. Slow crossing drives can satisfy  
781 neither approximation.

782 **The prediction.** The delay scaling is what this section supports as a pre-  
783 diction, and it needs no absolute calibration. Over the 6 uncensored legs the  
784 mean delay follows  $\Delta K \propto v^{0.444}$ , with a 95% slope interval of  $[0.394, 0.494]$ ,  
785 which does not contain  $1/2$ . What that shortfall measures is the escape cri-  
786 terion and not the scaling. Applied to a deterministic mean-field run seeded  
787 at the fluctuation floor, the same estimator returns 0.425 over the same legs  
788 at the same escape criterion, inside the interval on the measurement. That  
789 run has no noise to delay it, so the shortfall is the criterion's: it crosses the  
790 saturation between the floor it starts from and the level it is stopped at, and  
791 tightening the criterion to  $r \geq 0.05$  returns 0.490. The bias is reproduced  
792 where the answer is known, which is what licenses reading the measurement  
793 as the law rather than as a competing exponent.

794 Stated as a law rather than as a number: onset delay scales as the square  
795 root of the rate at which coupling crosses threshold. Testing it requires  
796 only that  $\dot{K}$  be monotonic in a manipulated rate—not its value, nor the  
797 installation factor  $\kappa$ , nor the field-to-coupling conversion rate  $\gamma$  of Eq. (15).  
798 It is therefore a within-subject design varying the rate of emergence from  
799 anaesthesia alone, each participant serving as their own control. Such a  
800 design must avoid the hazard this sweep exhibits: a fixed absolute escape

801 criterion meets a size-dependent critical fluctuation floor, so the criterion  
 802 has to be referred to each recording's own pre-threshold fluctuation scale  
 803 rather than to a shared number. The diffusion  $D$  is the remaining input,  
 804 and Section 8.1 notes that it is estimable from resolved phase traces through  
 805 the rate  $2D$  at which quadratic variation accumulates, which is a different  
 806 quantity from the concentration the propofol exercise could not range over.

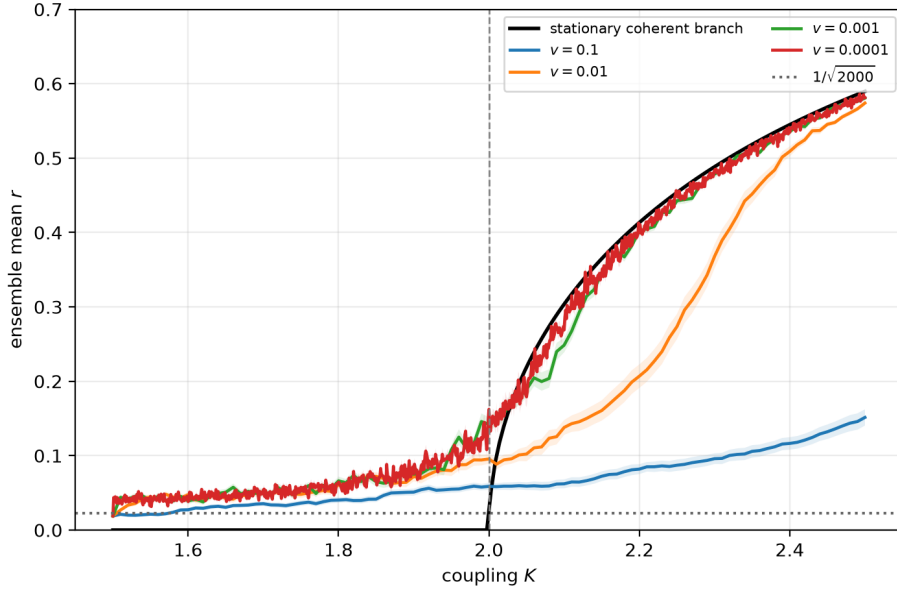


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds  $v$ , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

807 Additional controls address integration bias, population size and spatial  
 808 or signed coupling; the refined signed-network crossing bracket  $K_{\text{eff}}/D \in$   
 809  $[1.958, 2.009]$  contains 2.

### 8.3 A winding state leaves the order parameter ambiguous

The order parameter  $r$  is indifferent to where a phase sits in space. A field winding once across a periodic sheet therefore has  $r = 0$  while every patch of it stays locked. Such a field is not an artificial construction here. On a torus a winding is stationary for any isotropic symmetric kernel at identical frequencies: the coupling term is odd in the separation against a kernel even in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves this for any finite periodic array, this sheet included. A winding satisfies the positivity and identical-frequency hypotheses. It violates only the quarter-turn initial spread that the convergence result additionally requires before placing the limit at the uniform state.

On the sheet of Section 3's spatial control, a winding survives below an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4). That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same coupling, diffusion and kernel reaches  $r = 0.9657$  from a uniform start and  $r = 0.0083$  from a winding one. Both are stationary, with patch-local order 0.9015 and no winding defect. At fixed  $K$  the observable therefore takes two values spanning nearly its whole range. Coupling does not determine the resultant, and a low resultant does not by itself distinguish weak coupling from a winding field.

Under the quenched frequency spread of  $1.5 \text{ rad s}^{-1}$  that the spatial con-

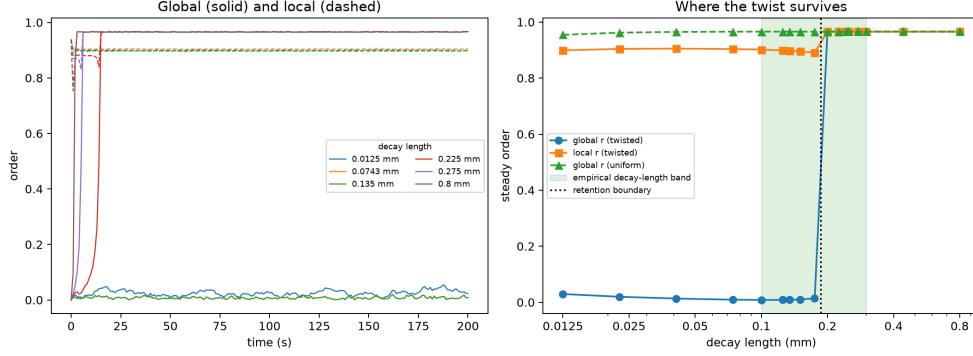


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of  $r$  at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

832 trol carries, the boundary moves to 0.1421 mm. The same comparison at  
 833 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.  
 834 Heterogeneity shortens the range that sustains a wave without removing it.

835 Which windings persist has a criterion behind it. Linearizing the phase  
 836 model about a winding returns a coupling of the same translation-invariant  
 837 form, each separation reweighted by the cosine of the phase the winding  
 838 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is  
 839 again even in the separation, so the same characters diagonalize it and ev-  
 840 ery mode’s rate is one of its eigenvalues. Where the winding advances by  
 841 at most a quarter turn across every separation the coupling reaches, the

842 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`  
 843 `stable_of_quarter_turn`); where some rate is positive, that mode grows  
 844 without bound (`charJacobian_growing_mode`). Moving coupling mass to  
 845 longer separations is what carries a phase advance past a quarter turn, and  
 846 that is the mechanism the decay-length sweep varies. The criterion bounds  
 847 rates rather than trajectories: asymptotic stability is not proved, so the  
 848 states that persist occupy a narrower range than it admits, and the bound-  
 849 ary quoted above is measured and not derived from it. The same kernel is the  
 850 potential's curvature, so a winding at  $r = 0$  is not the global minimum the  
 851 convergence result places its limit at, and one carrying a growing mode is not  
 852 a local minimum either: that result's quarter-turn hypothesis must exclude  
 853 a winding rather than merely decline to cover it (`char_not_potential_min`).

854 An amplitude layer decides which of these states exist, and leaves what  
 855 the resultant reports unchanged. Every oscillator of the phase model sits on  
 856 the unit circle, so a field with an amplitude node has no representation in  
 857 those dynamics. The real Stuart–Landau extension supplies one on the same  
 858 coupling, and leaves intact the gradient structure the convergence results rest  
 859 on. The state becomes  $z_i \in \mathbb{C}$  with  $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$ . Its  
 860 phase equation is the Kuramoto equation with each coupling weighted by a  
 861 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is  
 862 this one with the amplitude ratios set to one, rather than a separate object.

863 What the added degree of freedom decides is which windings exist at all.  
 864 On a periodic array of any dimension, this sheet included, a winding requires

865  $\Gamma + \lambda > 0$ , with  $\Gamma$  the layer's linear gain. Here  $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$  is  
 866 the eigenvalue the kernel supplies, summed over separations, with  $\Delta\theta_d$  the  
 867 phase the winding advances across  $d$  (`exists_charState_iff`; on the ring  
 868  $\Delta\theta_d = 2\pi qd/n$ ). For the nonnegative kernels used here  $\lambda \leq 0$ , and moving  
 869 coupling mass to longer separations lowers it further. A broad enough kernel  
 870 therefore removes the winding outright. The sweep above reaches that same  
 871 boundary by a different route: the phase model retains every winding at  
 872 every coupling, and loses this one under integration instead. The two layers  
 873 bound the same regime from opposite sides. A phase recording does not  
 874 separate a state the amplitude layer excludes from one the dynamics fails to  
 875 sustain.

876 Three limits bound the layer for a reader of the cortical literature. These  
 877 states are windings and not spirals, and on the sheet that is a theorem rather  
 878 than a shortage of dimensions. Equal modulus at every site puts the phase  
 879 circulation at exactly zero on every plaquette, and on a periodic sheet the  
 880 circulations of *any* field sum to zero, so a lone defect is not a state of this  
 881 geometry at all. Relaxational dynamics has no propagation speed, which is  
 882 the quantity a cortical wave measurement leads with. And  $|z|$  is a normal-  
 883 form radius, which a measured envelope is not; this work identifies the two  
 884 nowhere. Supplemental Material gives the sweep, its uniform control, the  
 885 amplitude layer, the standing wave it admits inside the band, the potential's  
 886 curvature at a winding and the defect theorems, together with the remaining  
 887 limits of both.

888 A vanishing resultant is the absence of global agreement and not of agree-  
 889 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-  
 890 ancy bound of  $2\sqrt{2}L|\sin \Delta\theta|$ , with  $\Delta\theta$  the phase advanced across the patch,  
 891 which falls to zero as the same winding is spread over more sites (`chord_le-`  
 892 `of_char_patch`), while the global form of the same bound is vacuous at  $r = 0$ .  
 893 Encoders shared on overlaps therefore still bound content disagreement in-  
 894 side a patch of a state that no global observable calls coherent. The winding  
 895 number itself stays unconnected to content. It is, however, a property of the  
 896 state rather than of the cover read on it. On a ring the whole turns dropped  
 897 by the winding's affine local lifts are zero at every site but the wrap, where  
 898 the residue resets, and they sum around the loop to  $q$ ; a nonzero  $q$  obstructs  
 899 every global real-valued phase lift, so one state at one coupling has resultant  
 900 exactly zero and degree  $q$  (`winding_degree_obstructs`). Reweighting the  
 901 sites moves the first quantity and redrawing the loop's intermediate transi-  
 902 tions leaves the second where it was. The scope is the ring's: this is a loop's  
 903 winding and not a spiral, the transitions are those of the affine lifts rather  
 904 than of an unwrapped measurement, and the integer is identified with no  
 905 cortical observable.

906 This qualifies what a phase trace can establish. An onset criterion stated  
 907 in  $r$  alone registers a change of basin as readily as a change of coupling.  
 908 The protocol of Section 9.3 therefore needs a spatial observable beside the  
 909 resultant.

## 910 8.4 Coherent adaptation can lose environmental align- 911 ment

912 An adaptive coupling rule might supply representational structure, but its  
913 objective must reward the relevant agreement. For symmetric coupling, pair-  
914 wise sine terms cancel in the population mean drift. Minimizing squared drift  
915 at that fixed mean therefore rewards equalizing drift across nodes. In the  
916 frequency-cluster control, this favors coupling between unlike-frequency clus-  
917 ters, whereas the diagnostic environmental template rewards coupling within  
918 clusters.

919 The formalization proves descent of the squared-drift objective for a spec-  
920 ified gradient flow at fixed phases. Its functional differs from thermodynamic  
921 entropy production: at a coherent stationary density the probability current  
922 vanishes while phase-averaged squared drift is positive. Supplemental Mate-  
923 rial gives the operator, its gradient and the conditions of the descent result.

924 The finite control co-evolves noisy phases and symmetric, nonnegative  
925 couplings at fixed total resource. Its template is a diagnostic covariance of  
926 latent signals, never a training input. Across three seeds, second-half order  
927 remains between 0.9167 and 0.9222, while the within-cluster to between-  
928 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).  
929 Random-update, permuted-gradient and frozen-kernel controls are reported  
930 in Supplemental Material.

931 The immediate post-update objective reduction disappears when aver-

932 aged over complete update intervals. A prespecified grid of 6 rate–cadence  
933 combinations contains no case meeting the 5% sustained-reduction criterion  
934 while retaining coherence and losing alignment. The finding is coherent adap-  
935 tation away from the specified structure, with no demonstrated sustained cost  
936 reduction.

937 Kernel alignment and overlap compatibility are different observables.  
938 This control addresses acquisition of its designated template under its par-  
939 ticular objective. Learning models can explicitly include data in an objective  
940 (Corberi et al., 2026); an account of agreement about the cup must specify  
941 the content its learning rule is meant to preserve.

## 942 **8.5 Content agreement needs independent decoders**

943 A neural test needs independently identified domains, local descriptions and  
944 restriction maps. Two decoders should estimate a shared content on over-  
945 lapping domains before their agreement is examined. Discrepancy requires  
946 a null under decoding noise and controls using deliberately compatible and  
947 incompatible examples. An implementation on constructed data separates  
948 these cases at  $AUC = 0.986$ , on a 95% interval of  $[0.980, 0.992]$  resampled by  
949 trial; the compatible construction scores 0.476 on an interval containing one  
950 half. Shared-prior shrinkage can nevertheless manufacture agreement, and  
951 regional bias can manufacture disagreement. Per-patch decoding diagnostics  
952 detect these failures; the compatibility statistic alone does not.

953 Phase agreement cannot replace that measurement. On constructed con-

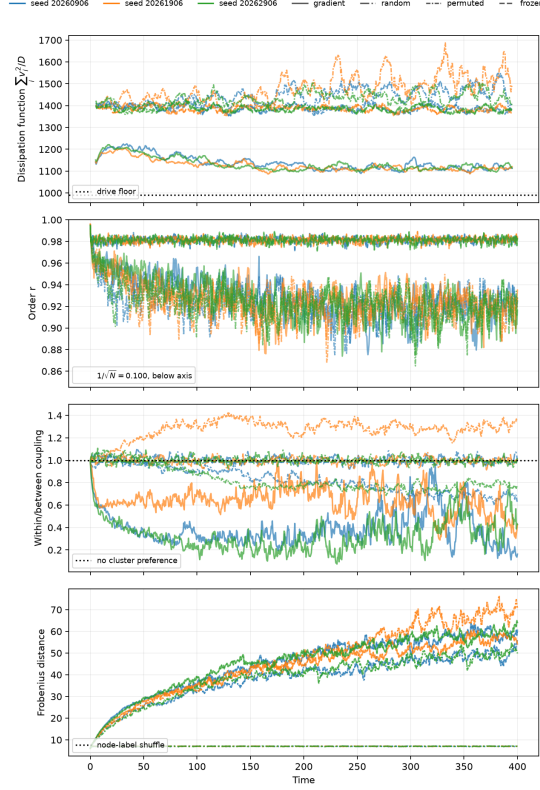


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

954 tent, phase carries a share 1.000 of the content entropy when content is built  
955 from it and 0.000 when it is not. A phase-derived statistic reports com-  
956 patibility on phase-locked patches whose local descriptions disagree. The  
957 implemented content statistic detects their incompatibility. These controls  
958 validate a measurement on known constructions; a neural content model and  
959 decoder remain to be established.

960 Choosing the cover also chooses a candidate subject: trivial descriptions  
961 can agree while carrying little content, and a partition can conceal disagree-  
962 ment. The useful choice must therefore be constrained by what neural ac-  
963 tivity encodes. The exact composition assumes compatibility of its supplied  
964 cover. The content mechanisms above identify additional hypotheses under  
965 which agreement can arise; they leave the empirical choice of content and  
966 the connection to that cover explicit.

## 967 **8.6 Estimator calibration on exploratory EEG**

968 The concentration range and the dependence structure decide whether the  
969 concentration-coherence relation can be tested at all. Under the specified  
970 log-density estimator and bias-floor criterion, discrimination at the observed  
971 concentration ceiling  $a = 0.542$  requires 3000 independent sites in the scanned  
972 designs, rising to 31000 when sites share a mean in groups of 10. These  
973 counts describe that estimator and dependence model; the intended recording  
974 requires its own calibration. Supplemental Material reports the bin-count  
975 controls and the full dependence-aware design.

976 That ceiling comes from exercising the estimators on public propofol-  
977 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-  
978 dows. Bipolar and common-reference montages produce different residuals.  
979 Bipolar-montage scalp EEG sits in the linear Bessel regime ( $a_{\max} = 0.542$ ,  
980 deviation from  $a/2 < 1\%$ ). The exercise confirms the estimator is well-  
981 behaved in this regime but does not test the nonlinear prediction. The  
982 bipolar results are compatible with the Bessel relation and with any mono-  
983 tone alternative in this regime. Temporal pooling changes the estimand, and  
984 sedation recordings leave the awakening hypothesis untested.

985 An informative dataset therefore needs sufficient concentration range or  
986 precision, calibrated estimators and an observation model for dependence,  
987 source mixing and reference choice. Testing the nonlinear prediction requires  
988 intracranial recordings (ECoG, sEEG or LFP) where local synchrony reaches  
989  $a > 1$ , or narrowband alpha-spindle burst analysis, at the sample sizes es-  
990 tablished above ( $\geq 31,000$  pooled phase samples or  $\geq 1000$  independent  
991 sites). Public mouse calcium imaging provides thousands of spatially located  
992 neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its  
993 7.65 Hz frame rate is below the phase-estimation band used here. Site count,  
994 state coverage and bandwidth are separate requirements.

## 995    **9    The cortical hypothesis and an awakening** 996        **test**

997    The cortical proposal identifies a physical source of coupling and asks whether  
998    its changes help explain recovery. The tests in Section 8 determine what  
999    observations must distinguish before a phase trace can support this identifi-  
1000   cation.

### 1001   **9.1   Physical identification and calibration**

1002   The cortical hypothesis identifies part of effective coupling with endogenous  
1003   extracellular electromagnetic interactions. Fields can modulate spike timing  
1004   (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models  
1005   propose coordination of distributed neural information (Pinotsis & Miller,  
1006   2022; McFadden, 2020). Phase organization under propofol motivates study-  
1007   ing this regime (Bastos et al., 2021), but does not measure coupling.

1008        The formal field predicate requires a jointly continuous kernel, a normal-  
1009   ized substrate, coupling equal to the kernel average, and no single site pro-  
1010   viding more than half that strength. Toy constructions satisfy it. A cortical  
1011   realization must specify the scale at which the kernel and its reduction ap-  
1012   ply: calcium imaging reports different state-dependent modularity patterns  
1013   at single-cell and mesoscale resolution (Kiyooka et al., 2026).

1014        Cortical columns are the obvious candidate unit, and they answer one  
1015   of the two demands the formalism makes. As the oscillator index they are

1016 a serious proposal: they are delineated by response properties and anatomy  
1017 rather than by the agreement they would be used to establish (Mountcastle,  
1018 1997), and a phase can be estimated from a column-sized territory at the  
1019 electrode spacings used for cortical wave measurements (Townsend et al.,  
1020 2015). As the content cover they are a poor one, for three separate reasons.  
1021 A parcellation into non-overlapping columns is a disjoint cover, which satisfies  
1022 the compatibility obligation without constraining anything (Section 4). The  
1023 cover condition additionally requires the patches to exhaust the substrate,  
1024 while much of association cortex has no accepted columnar parcellation. And  
1025 the column's standing as a canonical cortical unit is contested (Horton &  
1026 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore  
1027 supplies the oscillator index and leaves the content cover to be identified on  
1028 separate evidence, which is the bridge assumption E78 records.

1029 The spatial reduction is also testable. A row-sum aggregation at zero  
1030 diagonal tracks the uniform-coupling control over the tested spatial decay  
1031 range, while frequency heterogeneity displaces the operational threshold. The  
1032 fit window biases both estimates relative to  $2D$ ; agreement with the control  
1033 is therefore the supported conclusion. Supplemental Material reports the  
1034 reduction, its numerical error and the distinct assumptions needed for the  
1035 kernel-average identification in Eq. (5).

1036 The field commitment requires physical evidence beyond a choice of math-  
1037 ematical representation, and the available support comparisons do not supply  
1038 it. A finite matrix spread over positive-mass cells preserves its total cou-

1039 pling and phase correlation exactly, while point-supported vanishing under  
 1040 an atomless measure and a disadvantage from excluding a better-correlated  
 1041 pair concern specified support constraints; Section 6 reads the same compar-  
 1042 isons against a digital candidate. Neither supplies energy prices.

1043 A spike-time shift alone supplies no coupling rate. The dimensionless  
 1044 product of population size  $N$ , timing sensitivity  $\mathcal{T}$ , field amplitude  $E$  and  
 1045 frequency  $f$  needs a conversion rate  $\gamma$ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

1046 This rate describes how induced timing shifts accumulate into drift. Nei-  
 1047 ther it nor the installation factor  $\kappa$  follows from a heat bound. Testing the  
 1048 physical threshold requires independent estimates of the interaction kernel,  
 1049 its aggregation and diffusion in the same cortical regime.

## 1050 **9.2 Extracellular geometry and recovery**

1051 The proposed control variable is extracellular geometry. Extracellular vol-  
 1052 ume fraction is approximately  $\alpha = 0.2$  at tortuosity  $\lambda \approx 1.6$  (Syková &  
 1053 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie  
 1054 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively  
 1055 at  $0.3\text{--}0.6 \text{ S m}^{-1}$  (Barbour, 2017). At fixed transmembrane current density,  
 1056 an effective-medium description predicts that a larger extracellular fraction  
 1057 raises conductivity and reduces field amplitude.

1058 The coupling-gated hypothesis consequently predicts that extracellular  
 1059 contraction on awakening raises  $E$  and, holding the other factors in Eq. (15)  
 1060 fixed, raises effective coupling. This argument supplies a direction; the  
 1061 geometry–coupling relation, its magnitude and its role in human recovery  
 1062 require measurement.

1063 For a smooth crossing at nonzero speed, evaluating the stationary branch  
 1064 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

1065 This follows from Eq. (3) at fixed  $D$  and concerns the stationary branch  
 1066 evaluated at  $K(t)$ . Reading it as observed recovery requires an indepen-  
 1067 dently justified tracking approximation. The crossing controls in Section 8.2  
 1068 leave that approximation unsupported at onset, even though a separation of  
 1069 geometry and phase time scales can support it away from threshold. The  
 1070 awakening hypothesis therefore needs a calibrated dynamical prediction for  
 1071 an operational onset above the finite-size fluctuation floor. Crossing  $K_c$  alone  
 1072 does not supply a recovery latency.

### 1073 **9.3 A protocol for the awakening experiment**

1074 The physical test has six requirements:

- 1075 1. **Coupling and diffusion in the same units.** Estimate them from re-  
 1076 solved phase dynamics, separating heterogeneity and observation noise.

- 1077 Measure the drive or perturb the system to distinguish endogenous cou-  
1078 pling.
- 1079 **2. Geometry calibrated to coupling.** Validate the geometry observ-  
1080 able and its relation to effective coupling, with uncertainty. The con-  
1081 ductivity argument supplies only the expected direction.
- 1082 **3. Calibrated phase measurements.** Estimate  $r$  by resultant length  
1083 and  $a$  by unweighted regression of log density on centered cosine.  
1084 Choose site count and binning from a dependence-aware calibration  
1085 over the intended concentration range. The tested designs require  
1086 thousands of sites at the observed concentrations; temporal pooling  
1087 changes the spatial estimand.
- 1088 **4. An independent recovery endpoint.** Measure behavioural recovery  
1089 separately from phase organization.
- 1090 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-  
1091 old above the finite-size floor and evaluate its latency under the mea-  
1092 sured coupling trajectory, including ramp speed or fluctuations. Record  
1093 a spatial observable beside the resultant, since a winding field reports  
1094 a low resultant at coupling that sustains local order (Section 8.3).
- 1095 **6. Matched model comparison.** Compare coupling increase, diffusion  
1096 change, common drive and alternative recovery models under the same  
1097 observation and filtering assumptions.

1098 Failure of the concentration–coherence relation over an informative range  
1099 constrains the density model. Failure of the calibrated geometry–coupling  
1100 association constrains the physical gate. Better prediction of independent  
1101 recovery by a competing mechanism counts against its proposed role. A  
1102 favorable outcome still leaves the content of coordinated activity to be ex-  
1103 plained.

## 1104 10 Discussion

1105 Two things have been established here, and they are of different kinds. The  
1106 constraints of Section 2 hold outright: synchrony does not deliver compatible  
1107 content, and where shared encoders let it bound content at all, the bound  
1108 that holds uniformly over pairs does so only inside patches too small to  
1109 span the territories at issue, and distance costs either a connected cover or  
1110 a locked configuration with a declared spectral gap. The resultant does not  
1111 determine the state; a unique fixed point is not a reconstruction; a region that  
1112 cannot resolve a family cannot encode it; and agreement about a change takes  
1113 communication. Any account that reads unity off phase coherence has to  
1114 meet them, this one included. The account set out in Section 1 is conditional  
1115 at every step by contrast: each connection in it is a hypothesis supplied  
1116 rather than a result derived, and Table 1 names all eight. Its four non-  
1117 standard ingredients, named in Section 1, are the installed-energy condition,  
1118 the patch-local form of the content bound, the criterion separating a winding

1119 field from weak coupling, and the machine-checked composition with its joint  
1120 witness. What follows takes the conditions in turn, as commitments the  
1121 proposal incurs, as measurements that would bear on them, and as questions  
1122 the framework leaves open.

## 1123 **10.1 Unity and a minimal self as empirical commit-** 1124 **ments**

1125 We propose the state glued from compatible local descriptions as a corre-  
1126 late of conscious unity, and its accurate internal reconstruction as a correlate  
1127 of a minimal self. The motivations are that different contents belong to  
1128 one experienced situation and that the situation is experienced from a per-  
1129 spective. The conditional results identify relations a realization would need:  
1130 sufficient installed resources for the specified phase model, agreement about  
1131 shared content, and an encoding that supports reconstruction at the required  
1132 fidelity. Discharging E12–E89 for cortex would make the composition a state-  
1133 ment about cortex: that a cortical field above threshold relaxes to a cover  
1134 whose compatible local descriptions glue, and that the glued state is the  
1135 unique fixed point of a reconstruction map realized there. It would leave the  
1136 two identifications above exactly where they are, because neither is among  
1137 the eight and neither follows from their conjunction; they are the commit-  
1138 ment this subsection names, and Section 10.2 lists the three observations  
1139 that would count against it.

1140        Compatibility allows different descriptions to contribute different infor-  
1141        mation. It also permits uncertainty and conflicting judgments: apparent cup  
1142        position and judged cup position can be distinct quantities in a consistent  
1143        description of their discrepancy. Which quantities are shared must be de-  
1144        termined independently. Choosing them solely to secure agreement would  
1145        deprive the proposal of empirical force.

1146        The minimal-self proposal concerns the perceiver’s position and condition  
1147        within the situation, such as the cup’s location relative to a body that can  
1148        reach it. The encoding could be implicit and need carry no autobiograph-  
1149        ical narrative. Its proposed necessity is philosophical: even an unreflective  
1150        episode has a perspective. Gluing by itself requires no self-encoding, and re-  
1151        constructability alone supplies no functional role. Altering the encoded hand  
1152        or viewpoint while holding the external scene fixed should change estimates  
1153        of the perceiver’s relation to the cup as the readout predicts. Whether that  
1154        represented perspective is experienced remains a further question.

## 1155    **10.2    What a test of the full proposal must establish**

1156        Both uniqueness results concern specified objects. They choose neither the  
1157        cover nor the encoding, and do not select between candidate subjects. In-  
1158        tegrated information theory treats exclusion explicitly (Tononi et al., 2016);  
1159        here selection requires independent evidence about neural organization. A  
1160        constant readout can contract and fix a predetermined state while discard-  
1161        ing its input. The family criterion in Section 5.2 tests the sensitivity that

1162 this construction lacks, provided the relevant states and fidelity are declared  
1163 independently.

1164 A test must separately measure coupling relative to diffusion, content  
1165 compatibility and self-reconstruction, alongside an independent endpoint for  
1166 experience. Meeting all three without a unified episode would challenge suffi-  
1167 ciency. A unified episode with incompatible descriptions would challenge the  
1168 unity identification; one lacking the specified self-encoding would challenge  
1169 the reflexive requirement. Awakening, dreaming and dissociative states offer  
1170 possible settings, but absence of report cannot by itself define absence of  
1171 experience.

1172 The next physical requirement is a calibrated geometry–coupling relation  
1173 and resolved phase dynamics. The installation bound contributes a nec-  
1174 essary resource condition where its hardware representation applies; cortical  
1175 mode prices and field-mediated coupling require measurement. For the repre-  
1176 sentational connection, an informative experiment asks whether a controlled  
1177 change in coordination alters independently decoded content agreement, with  
1178 controls for sensory information and decoder quality. The constructed-data  
1179 estimator establishes feasibility for specified contents. Neural local descrip-  
1180 tions, shared overlap encoders and the self-encoding have yet to be identi-  
1181 fied. The population-size dependence of the phase-to-content bound limits  
1182 its quantitative use in the form Section 4.2 reads off the sheet, and each  
1183 way around it asks the measurement for something further: a located excep-  
1184 tional set, before a count of disagreeing pairs bears on a named overlap; a

cover whose nerve is connected and whose patch diameters are measured; or  
a frequency-locked population over a coupling graph whose connectivity is  
measured structurally rather than estimated from the activity the bound is  
about.

The separation results explain why these measurements must be independent. A stationary distribution leaves several recovery mechanisms indistinguishable. Equal phase order can accompany incompatible descriptions or different task information. The tested plasticity rule preserves coherence while losing its designated template. Each finding concerns the specified model and observable; their role is to delimit the evidence needed for the proposed connections.

### 10.3 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. That permission is wider than the unconditional results of Section 2 alone. The conditional chain is substrate-neutral in the same way: its field condition asks for a jointly continuous kernel on a normalized substrate, a coupling equal to that kernel’s double average, a noise that is the field’s own, and

no single site carrying more than half the coupling—and none of these distinguishes an electromagnetic kernel from any other continuous mean-field kernel. The site condition is the only one with physical flavour, and it holds vacuously wherever the substrate has no atoms (`no_site_dominates`), which is the continuum case the electromagnetic hypothesis is about. The chain therefore transfers to extracellular ion diffusion, to gap junctions modelled as a density, or to a mechanical or optical medium, modulo the calibration of Section 9.1. What carries the electromagnetic identification is that calibration together with the extracellular-geometry argument of Section 9, which is the one place a field equation enters. The formalization checks implications under their hypotheses, and its witnesses establish realizability on declared models. Two things stand outside that result and are not supplied by it: the eight hypotheses are discharged for toy components rather than for cortex, and the identification of the glued state with unity and of its reconstruction with a minimal self is a further commitment that the eight do not contain.

The explanatory question is whether the specified relations occur in neural activity and matter for experience. In the cup example, this means establishing how coordination helps maintain one represented situation and how the perceiver’s place within it contributes to the episode. The proposal identifies candidate correlates and tests of that identification. Why those relations would be accompanied by experience remains unexplained.

## Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

| Symbol                    | Defined     | Meaning                                                                                                   |
|---------------------------|-------------|-----------------------------------------------------------------------------------------------------------|
| $\theta_i, \omega_i$      | Eq. (1)     | Phase and natural frequency of oscillator $i$                                                             |
| $N$                       | Eq. (1)     | Number of oscillators in the population the quantity is read on                                           |
| $D$                       | Eq. (1)     | Phase diffusion, in inverse time                                                                          |
| $K_{ij}, K$               | Eq. (1)     | Pairwise and scalar mean-field coupling, $K_{ij} = K/N$                                                   |
| $K_c$                     | Eq. (2)     | Synchronization threshold, $K_c = 2D$                                                                     |
| $r, r_N$                  | Eq. (1)     | Resultant length (order parameter), on the population it is read on                                       |
| $a, \psi$                 | Eq. (2)     | Concentration $a = Kr/D$ and collective phase of the stationary density                                   |
| $q$                       | Section 8.3 | Winding number of a phase field read around a ring                                                        |
| $\mu$                     | Eq. (4)     | Substrate measure, normalized to one                                                                      |
| $\Xi, \phi_i, u_i, c_i$   | Eq. (4)     | Actuator configuration; mode profile, installation price and occupancy of mode $i$                        |
| $U_{\text{inst}}, \kappa$ | Eq. (5)     | Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$ |
| $L$                       | Section 4.2 | Lipschitz constant of an encoder or readout                                                               |
| $\varepsilon, \delta$     | Section 4.2 | Content and observation discrepancy tolerances                                                            |
| $M, b$                    | Eq. (10)    | Size of a declared family of states and code length in bits                                               |
| $\Lambda, \tau$           | Eq. (8)     | Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time                                  |
| $T$                       | Section 5   | Number of communication rounds to a deadline                                                              |
| $\nu$                     | Section 6   | Fan-in of a communication graph: the most sites any one site hears from                                   |
| $k_B T, B$                | Eq. (14)    | Thermal scale, read as one symbol, and heat budget of a register                                          |

Table 2 (continued)

| Symbol                   | Defined     | Meaning                                                                                                        |
|--------------------------|-------------|----------------------------------------------------------------------------------------------------------------|
| $\gamma, \mathcal{T}, f$ | Eq. (15)    | Field-to-coupling conversion rate, timing sensitivity and frequency                                            |
| $\alpha$                 | Section 9   | Extracellular volume fraction                                                                                  |
| $\lambda$                | §8.3, §9    | Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity                                 |
| $\Omega, \lambda_2$      | Section 4.2 | Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph |
| $\beta, k$               | Section 4.2 | Chord diameter of a patch and the diameter of a cover's nerve, in hops                                         |
| $\mathcal{C}, e_{i,x}$   | Section 4.2 | Content space and the phase encoder patch $i$ reads at site $x$                                                |
| $E, R$                   | Eq. (10)    | Encoder and readout of the reconstruction criterion                                                            |
| $\mathcal{A}, F$         | Eq. (8)     | Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$                           |
| $\mathcal{E}$            | Eq. (13)    | Energy observable shared by the stages of a continuing protocol                                                |
| $z_i, \Gamma, A_{ij}$    | Section 8.3 | Complex amplitude, linear gain and coupling matrix of the amplitude layer                                      |
| $\Delta\theta_d$         | Section 8.3 | Phase a winding advances across separation $d$                                                                 |
| $E$                      | Eq. (15)    | Field amplitude, in the cortical identification                                                                |
| $U$                      | §3.2, §4    | Stored energy $U(\Xi)$ of a configuration; a patch $U$ of the cover                                            |
| $s, s_*$                 | §4–5        | A global section and the one the reconstruction fixes                                                          |

## 1229 Data and code availability

1230 The Lean development, simulation scripts, tests and saved numerical sum-  
1231 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`  
1232 `cortical_coherence`. Publication simulation values are generated from

1233 saved summaries without rerunning production sweeps. The exploratory  
1234 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-  
1235 processing and selection are described in Supplemental Material. The repos-  
1236 itory contains the reproducible analysis and does not redistribute the raw  
1237 recordings.

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1239 This research received no external funding. The author declares no compet-  
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## 1241 **Use of AI tools**

1242 AI assistance to this work was substantial and is not separable by task. Sev-  
1243 eral coding assistants and chat interfaces were used in an interleaved way  
1244 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,  
1245 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,  
1246 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-  
1247 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,  
1248 accessed with the Hermes CLI. Between them they contributed to the Lean  
1249 development, the simulation and analysis code, and the drafting and revision  
1250 of both documents. Because the tools were used in combination and in alter-  
1251 nation, no section, proof, figure or result is attributable to any one of them,

1252 and none is claimed to be. The author is responsible for all content, including  
1253 everything a tool produced. Machine checking establishes the stated Lean  
1254 results under their hypotheses; it does not validate biological interpretations  
1255 or replace scientific review of generated code and prose.

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